B.Tech. Degree VII Semester Examination, November 2009

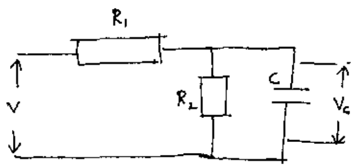
EB/EC/EI 705 (D) MECHATRONICS
(2006 Scheme)

Time: 3 Hours Maximum Marks: 100

PART – A
(Answer ***ALL*** questions)

(8 x 5 = 40)

I. (a) Describe the working of an absolute optical encoder.
(b) Explain the problems of loading when a measurement system is being assembled from a sensor, signal conditioner and display.
(c) With the help of a neat sketch explain the principle of a pilot operated valve.
(d) Explain the operation of variable reluctance stepper motor.
(e) Derive the relation slip between the output, the potential difference across the capacitor C of Vc, and the input V for the circuit shown below.



(f) Explain the working of a MEMS accelerometer.
(g) Compare P, P-I and P-I-D as modes of process control.
(h) Explain point-to-point and continuous path control modes of robots.

PART – B
(All questions carry ***EQUAL*** marks)

(4 x 15 = 60)

II. (a) Describe the key elements of a Mechatronic system.
(b) List out the factors to be considered in selecting a sensor.

OR

III. (a) List the characteristics of an ideal op-amp.
(b) Explain the operation of a logarithmic amplifier.

IV. Describe in detail the various solid state devices that can be used to electronically switch circuits.

OR

V. (a) Describe the working of a quick return mechanism.
(b) Classify journal bearings based on the lubricant used.

VI. (a) Derive the differential equations for a permanent magnet dc motor.
(b) Define the following performance measures for a second order system:
(i) Rise time
(ii) Peak time
(iii) Overshoot
(iv) Settling time

OR

VII. Explain in detail the various machining processes to fabricate MEMS.

VIII. (a) With the help of a block diagram, explain a programmable logic controller. Elaborate on the functions of each block.
(b) Describe the various PLC programming languages.

OR

IX. (a) Explain the basic components of Flexible Manufacturing System (FMS).
(b) List out the benefits that can be expected from an FMS.

****